Day 21

Router_Multicast

Router:

* What is Router? To route the data to one branch Based on a Condition. Suppose you have multiple Conditions then you can use iRouter

Example: Scenario: postman → Body → raw → xml →

Constraint <Status> active </Status) → then we need Send it to the one direction or if status is not 'active' the we need to send it to another route

At, this above Scenario we will use Router Based on a Condition ✓

Demo did in System

** If you are using Router → you must define the default route

Default route: CPI → middleware tool → process Flow → Step-by-step process → if process start → must be end- Some where ✓

→ functionality of CPI

→ Suppose it process starts → it should be end properly ~

Q) if you recive status other than active and inactive then what will happen? Now what CPI has to do at that time?

So to over come this situation while using the Router pallete function then you must and should define the default & Route ~

* This condition not only works with the xml but if we have non xml conditions also it will work

→ non → xml Conditions like ⇒ property (or) headers

* usage of operators in non xml Conditions

=, !=, >, >=, <, <=, and, or, Contains,

not contains, in, not in, regex, not regex

all the operations will work for router

${header.sender} = 'i'

↳ all the above operations will work here ✓

* Invalid format of Condition expression value.

↓

error (when you get this error)

What it is ment by?)

→ if we provide any wrong format in Non xml format in Condition then we will get that error → because we provide "value" in single Quotes for example ${header.Routerid} = '1'

Router: processing:

Expression Type: xml (or) Non xml

Condition: { // ---- } (or) ${ ---- }
xpath (xml) non xml

Default Route:

... from interview perspective it is very important → why?

① What is Router → to send the data in 1 Branch Based on a Condition ✓

② What if multiple Condions are satisfied →
let's say i have 3 routing Conditions and one default route ✓

→ Let's say all 3 Conditions are satisfied
then where the data will go ?

Scenario. Body → <STATUS> Active </STATUS>

Important ✓ header → RouterId > 1 (4) eg

then what will happen two Conditions are Satisfied ✓

⇒ So here → how many Conditions are satisfied that's not a matter →

Message never go to multiple Branches it only route the data to the one Branch ✓

irrespective of how many conditions are matched or not matched → that is not relevant.

Now → another → Pallete function:

* **Multicast** : Suppose if you want to send the data to the multiple Recievers at the same time then we will use multicast pallete function.

eg: Requirement is → we have → 5 recievers & → if we need to send the same data to that 5 recievers at the same time (without any condition) → we use **Multicast**

Multicast → **Same data** → more than one reciever.

→ Send → Same time

Now, Multicast

- Parallel Multicast
- Sequential Multicast

Sequential multicast: if i want to send the data in sequence order (one by one) then we will use Sequence multicast

eg: How you can define the sequential

Routing Sequence → move up
move down

↓

filter

(Q) If you get any failure in any one flow then what happen to the second Sequential flow → whether it will be triggered or no ?

Realtime Scenario: → multiple Recievers → Multicast (parallel)

→ first receiver → and then Second receiver (Sequential)

* parallel & Sequential you have to see whether its depends on the requirement

→ multiple recievers → reciever says → no dependency

→ you → call → multiple recievers at the same time → parallel multicast

(or)

→ 1st receiver → depends → 2nd receiver

→ Sequential multicast →

* When we are talking about multicast Two Questions are very important

① In Sequential multicast, what happens

ⓐ if 1st receiver fails, means -

→ will data goes to second receiver "not" go to second receiver = no

ⓑ : In parallel multicast, what happens if 1st receiver fails, means

→ Will data goes to second receiver

It will go to second receiver

yes ✓

② In Sequential multicast, what if i created a header/property ~~if body~~

in Branch (1) can i use same header & /property in Branch (2) ?

Ans : NO

(b) In parallel multicast, what if i created a header/property in branch (1), can i use the same headers or properties in branch 2 ? (POC) → is it yes (or) no

Ans : NO

* payload → 2 records → EMPID 4 & EMPID 5 → Status → for '4' → Active & Status → for '5' →

In Active

↓

Router cannot handle the above situation if we send more than 1 record